

GitHub, Docker and Docker Hub Task Guide (15 Steps)

Step 1: Fork the Repository

Open the GitHub repository and click Fork to create a copy under your GitHub account.

Step 2: Clone Your Fork

```
git clone https://github.com//cicdakshat.git
cd cicdakshat
```

Step 3: Create a Branch

```
git checkout -b feature-branch
```

Step 4: Create a File

```
touch myfile.txt
echo 'This is my first commit' > myfile.txt
```

Step 5: Commit Changes

```
git add .
git commit -m 'Added myfile.txt'
```

Step 6: Push Branch to GitHub

```
git push -u origin feature-branch
```

Step 7: Raise and Merge Pull Request

Create a Pull Request in GitHub and merge it into the master/main branch.

Step 8: Create Dockerfile

```
Example Dockerfile:
FROM openjdk:8
EXPOSE 80
COPY target/devops-integration.jar devops-integration.jar
ENTRYPOINT ["java","-jar","/devops-integration.jar"]
```

Step 9: Build Maven Project

```
mvn clean package
Verify target/devops-integration.jar exists.
```

Step 10: Build Docker Image

```
docker build -t devops-integration:v1 .
```

Step 11: Verify Image

```
docker images
```

Step 12: Run Container

```
docker run -d -p 8080:8080 --name devops-app devops-integration:v1
```

Step 13: Login to Docker Hub

```
docker login
```

Step 14: Tag Docker Image

```
docker tag devops-integration:v1 /devops-integration:v1
```

Step 15: Push Image to Docker Hub

```
docker push /devops-integration:v1
```

Simple Text Docker Image Example

Dockerfile:

```
FROM alpine:latest
```

```
CMD echo "Hello, this is my custom Docker image!"
```

Build:

```
docker build -t text-image:v1 .
```

Run:

```
docker run text-image:v1
```